LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Electricity

Formula sheet

Quantity	Formula	SI unit
Current	$I = Q/t$	ampere (A)
Potential difference	$V = W/Q$	volt (V)
Ohm's law	$V = IR$	ohm (Ω)
Resistance of a wire	$R = \rho L/A$	ρ in ohm metre (Ω m)
Series	$R_s = R_1 + R_2 + R_3$	same current through all
Parallel	$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$	same voltage across all
Heating (Joule's law)	$H = I^2Rt$	joule (J)
Power	$P = VI = I^2R = V^2/R$	watt (W)
Energy	$E = Pt$	$1\ \text{kWh} = 3.6 \times 10^6$ J

Must-know points

- Charge on an electron = 1.6×10^{-19} C. Conventional current flows opposite to electron flow.
- Ammeter goes **in series** (low resistance). Voltmeter goes **in parallel** (high resistance).
- Ohm's law holds at constant temperature: the V–I graph is a straight line through the origin, and its slope gives R.
- R depends on length ($R \propto L$), area ($R \propto 1/A$), material (ρ) and temperature. Doubling the length doubles R. Doubling the diameter makes R one-fourth.
- Good conductors (Cu, Al) have low ρ. Alloys (nichrome) have high ρ and don't oxidise easily, so they are used in heaters, toasters and irons.
- In parallel, R_p is smaller than the smallest resistor. A wire cut into n equal parts joined in parallel gives R/n^2.

Why homes use parallel wiring: each appliance gets the full 220 V, can be switched on or off separately, and keeps working if another fails.
Uses of heating effect: bulb filament (tungsten, very high melting point), electric iron, toaster, heater, fuse (low melting point wire).

Worked examples

- 5 Ω, 10 Ω, 30 Ω in parallel: $1/R = 1/5 + 1/10 + 1/30 = 10/30$, so R = 3 Ω. At 12 V: I = 4 A.
- Bulb 220 V, 100 W: $R = V^2/P = 484$ Ω. On 110 V: $P = 110^2/484 = 25$ W (one-fourth).
- Fridge 400 W × 8 h × 30 days = 96 kWh. At ₹3 per unit, the cost is ₹288.

Common mistakes

- Not converting units: mm → m ($0.5\ \text{mm} = 0.5 \times 10^{-3}$ m), W → kW, minutes → seconds.
- Using diameter instead of radius in $A = \pi r^2$.
- Writing 'units' of energy without saying 1 unit = 1 kWh.

Quick check

1. Two 6 Ω resistors are in parallel. What is the equivalent resistance?

2. How much current does a 60 W, 220 V bulb draw?